

UBT Scintillator Digitization

Geometry, Optical Model, and ADC Digitizer Validation for the SHiP Upstream Background Tagger

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Outline

- 1 Detector Definition
- 2 Simulation & Digitizer Results
- 3 FairShip Digitization & Expected Rates
- 4 Conclusions & Next Steps

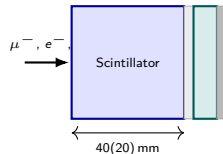
Geometry & Sensor Stack

Volumes

- World: Geant4 air box, half-size (0.5, 0.5, 0.5) m.
- Default tile: $40 \times 40 \times 10$ mm³.
- UBT_SCINTILLATOR_SIZE_MM=20 selects $20 \times 20 \times 5$ mm³; other sizes rejected.
- Sensor active face fixed at 6×6 mm² for both tile sizes.

Coordinate convention

- Scintillator center at (0, 0, 0); beam along +z; sensor stack on the +z face.
- Muon source plane upstream at negative z.
- Gun default: μ^- at (0, 0, -30 mm), direction (0, 0, 1); kinetic energy uniform in 1–20 GeV.



p -terphenyl+POPOP scintillator $\rightarrow$ grease (0.2 mm) $\rightarrow$ glass window (1.0 mm) $\rightarrow$ photocathode/SiPM (0.1 mm).

Parameter	40 mm tile	20 mm tile
Transverse size	40×40	20×20
Thickness	10 mm	5 mm
Sensor face	6×6 mm ²	
Beam-spot half-size a	20 mm (init.)	≤ 10 mm

Materials, Optics & Timing Model

Materials

- Scintillator: effective para-terphenyl + POPOP, $\rho = 1.23 \text{ g/cm}^3$; bulk composition $\text{C}_{18}\text{H}_{14}$. POPOP enters via the emission spectrum, not as an explicit dopant fraction.
- Window: Pyrex glass Grease: hydrocarbon
Photocathode: silicon slab World: air.

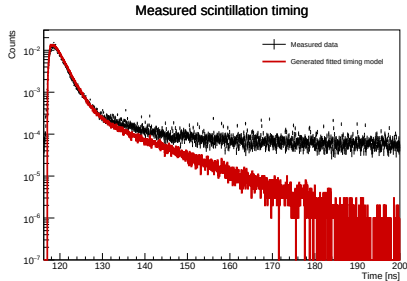
Photon detection chain

- Two Bernoulli filters at the photocathode boundary: flat light-collection $\epsilon_{\text{LC}} = 0.30$, then energy-dependent $\epsilon_{\text{PDE}}(E_\gamma)$.
- Reference: Hamamatsu S13360-6050PE ($6 \times 6 \text{ mm}^2$, $50 \mu\text{m}$ pitch); $\epsilon_{\text{PDE}}(450 \text{ nm}) = 40\%$.
- Digitizer gain $G = 2.8 \times 10^6$; $\text{ADC} = \text{round}(Q[\text{pC}])$ – one pC equals one ADC count.

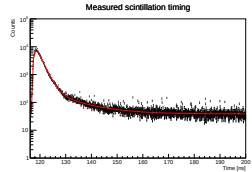
Timing constants	
τ_{rise}	0.7 ns
τ_{fast}	2.0348 ns
τ_{slow}	11.3607 ns
Y_{fast}	0.889145
Y_{slow}	0.110855
Total yield	10 000 γ/MeV

Shared emission spectrum (fast *and* slow components) peaks at 2.93 eV (relative intensity 1.00), spanning 2.75–3.10 eV – the two components differ only in time structure and relative yield, not spectral shape.

Scintillation Timing Model Validation



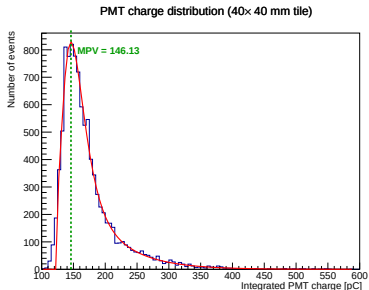
Normalized measured spectrum compared with times generated from the fitted two-component model.



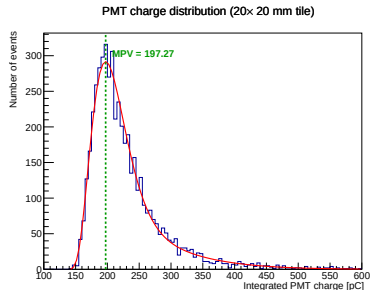
Input: measured spectrum over the fit interval.

One million times are generated from the fitted common-rise, fast-decay, slow-decay components (fixed seed). Both histograms are normalized, so the overlay tests timing *shape* only.

Simulated Integrated PMT-Charge Distributions



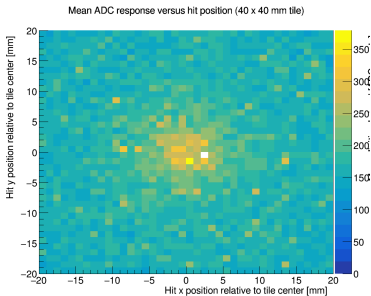
$40 \times 40 \text{ mm}^2$ tile



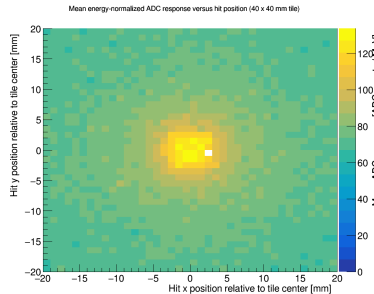
$20 \times 20 \text{ mm}^2$ tile

Simulated integrated PMT-charge distributions in picocoulombs for the two tile sizes. Numerically, one pC corresponds to one ADC count in the present standalone conversion, but the plotted observable and horizontal axis are PMT charge in pC.

Spatial Uniformity of the ADC Response



Raw mean ADC vs. position

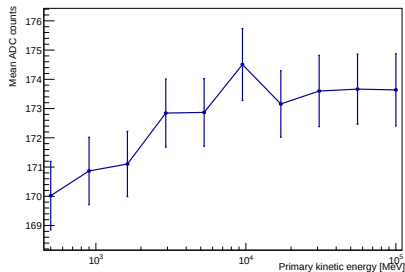


ADC/MeV, energy-normalized

Spatial uniformity map of the raw response and energy-normalized map for tiles $40 \times 40 \text{ mm}^2$. Empty bins mark regions without selected events, not zero response.

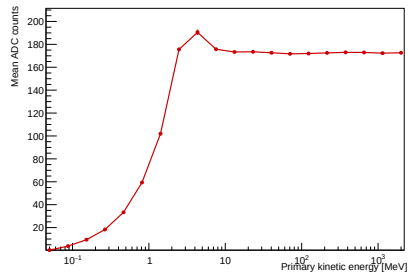
Individual Per-Particle Mean-ADC-vs-Energy Curves

Mean ADC Counts vs Primary Energy: Muon



μ^-

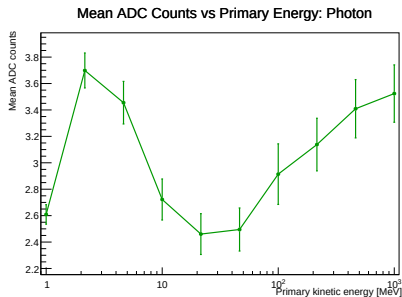
Mean ADC Counts vs Primary Energy: Electron



e^-

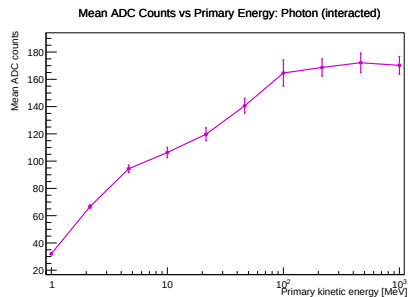
Muon response is essentially flat (≈ 170 ADC counts). The electron scan is peaking ≈ 190 ADC at 4.3 MeV, settling onto the 172–174-count plateau above 13 MeV.

Photon Response, Split by Interaction



All γ events, interacted or not

The all-events curve is dominated by the low interaction probability in the thin tile and stays near zero. Restricting to non-zero deposits reveals a rise.



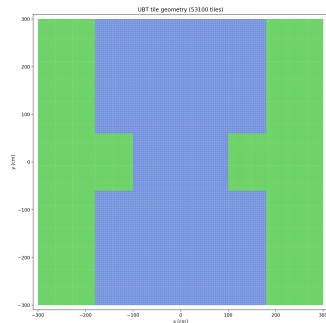
Only interacted γ events

Two Levels of the Final UBT Mapping



Coarse transport map: 225 regions of $40 \times 40 \text{ cm}^2$

Only the coarse regions (left) are physical FairShip geometry volumes; the small tiles (right) are inferred mathematically during digitization. Blue: $4 \times 4 \text{ cm}^2$ tiles; green: $2 \times 2 \text{ cm}^2$ tiles (three-column full-height sides, two-column-per-side central neck).

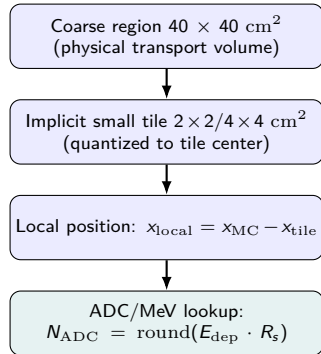


Constituent-tile map for digitized hit coordinates

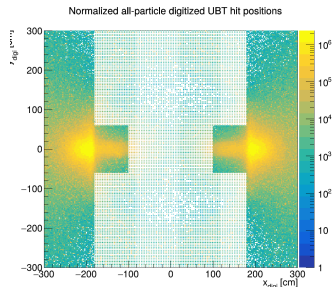
FairShip Digitization: Coarse Plane to Small-Tile ADC

From physical volumes to a small-tile response, in short:

- 1 Transport only sees **coarse** regions: a 15×15 grid of 225 sensitive $40 \times 40 \text{ cm}^2$ volumes, each locked to a 2 cm or 4 cm constituent-tile pitch (102 vs. 123 regions).
- 2 Every MC point is **requantized** onto the implicit small tile: $i_q = \lfloor (q_{\text{MC}} - o)/s(D) \rfloor$ picks the tile index; the digitized position becomes that *small tile's center*, not the parent region's.
- 3 ADC uses the point's **exact local position** inside the small tile, $x_{\text{local}} = x_{\text{MC}} - x_{\text{tile}}$, looked up in a $1 \times 1 \text{ mm}^2$ ADC/MeV map from the standalone Geant4 study.
- 4 $N_{\text{ADC}} = \text{round}(E_{\text{dep}}[\text{MeV}] \cdot R_s(x_{\text{local}}, y_{\text{local}}))$; triggered when $N_{\text{ADC}} \geq 100$.

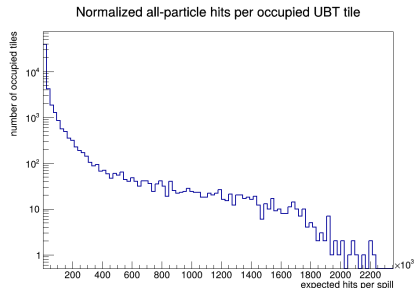


Expected Particle Rate per Tile



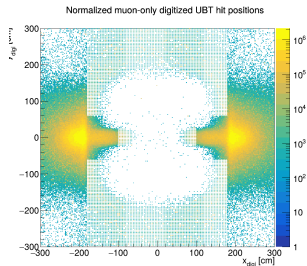
General spatial rate map, all species (weighted, log z)

Digitized UpstreamTaggerHit rates over the full detector plane from the MinBias2026 muon-background sample, weighted to $N_{\text{total}} = 4.12 \times 10^8$ muon-background events per spill.

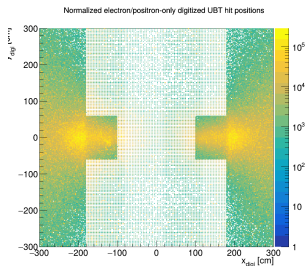


Per-tile expected hit-rate distribution (log y)

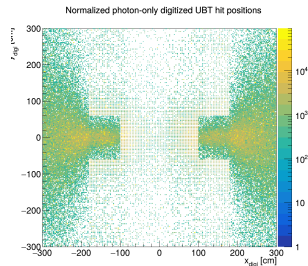
Expected Rate per Particle Type



μ^- only



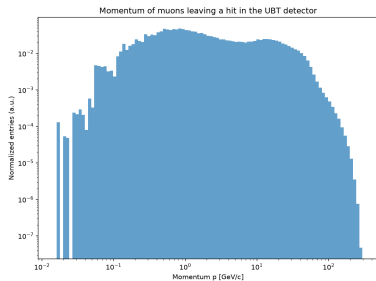
e^-/e^+ only



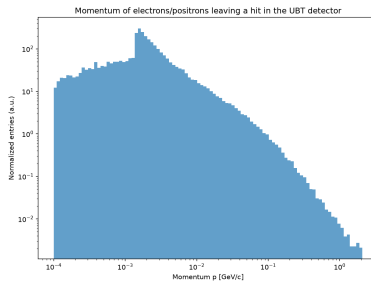
γ only

Same weighted spatial rate map as the previous slide, split by MC primary species. Muons dominate the rate across the whole plane; electrons/positrons and photons are comparatively more concentrated close to the beam axis, consistent with their softer momentum spectra.

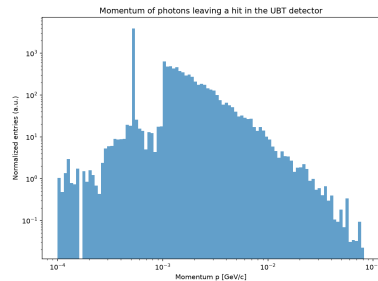
Momentum Distribution of Each Particle



μ^- : up to a few hundred GeV/c



e^- : $\mathcal{O}(1)$ GeV/c



γ : $\mathcal{O}(10)$ MeV/c

Momentum of particles leaving an UpstreamTaggerPoint.

Conclusions

- The standalone Geant4 model **reproduces the measured scintillation timing shape** (shared emission spectrum, common-rise/fast/slow components) and gives consistent simulated PMT-charge distributions for both the 40×40 and 20×20 mm² tiles.
- The ADC response is **spatially uniform** once energy-normalized, and per-particle ADC-vs-energy calibration curves are established for muons, electrons/positrons, and interacted photons.
- The FairShip digitizer correctly **requantizes** MC points from the coarse 40×40 cm² transport regions onto the physical $2 \times 2/4 \times 4$ cm² tile grid, applying each tile's local ADC/MeV response.
- The digitization model gives a **more reliable per-tile rate estimate** based on each hit's position in the UBT plane.

Next Steps

- Re-validate the expected per-tile rate map, evaluate the number of tiles with rate about 1 MHz threshold.
- Address the timing model: the standalone simulated timing does not yet match real tile measurements; FairShip currently substitutes a real-data-based timing smearing in its place.
- Extend the ADC calibration and rate study to additional energies and angles, and cross-check against real UBT tile test-beam data once available.
- Using the ADC-vs-momentum and momentum-distribution results, study a possible rate reduction from an ADC threshold cut.